Biosensors for Probiotic Bacterial Detection

MSc Biotechnology

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Background & Rationale

Why this matters: Probiotic function is often **strain-dependent**, but **species-level detection** remains vital for screening, QC, and process monitoring. *L. agilis* is promising yet under-profiled and frequently misidentified by conventional methods
Gap: Culture counts, MALDI-TOF, and metagenomics each have blind spots (VBNC cells, library coverage, cost/complexity). Rapid, low-cost, species-specific tools are scarce for *L. agilis*.
Approach: A **two-tier platform**—(1) **visual colorimetric biosensor** (natural pH-sensitive dyes) for instant readout of metabolic activity; (2) **CRISPR-Cas12a** confirmation targeting the 16S–23S ISR for species specificity.
Aim: Build and validate a deployable *L. agilis* assay that pairs colorimetry with CRISPR-Cas12a for rapid, species-level detection.

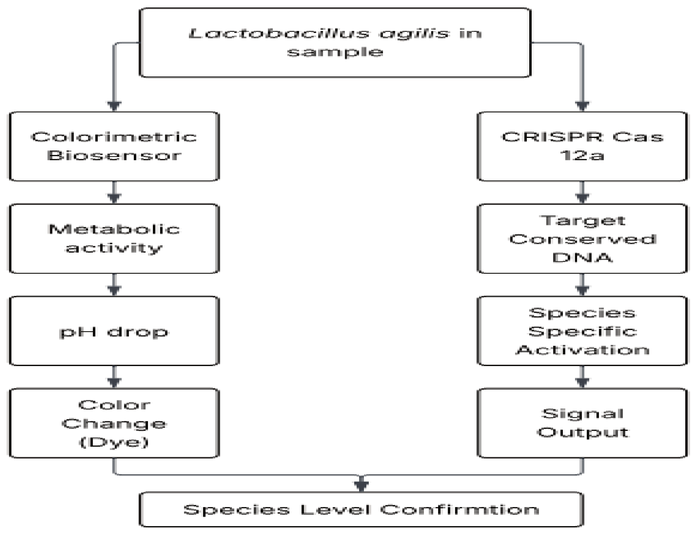

Figure 1: Schematic Overview of the L. agilis Detection Platform

1 Standardizing viable cell counts (OD↔CFU) for reproducible inputs

- Defined anaerobic growth; OD range **0.1–1.1** showed linear OD↔CFU
- Provides a **reliable baseline for bioink inoculation**

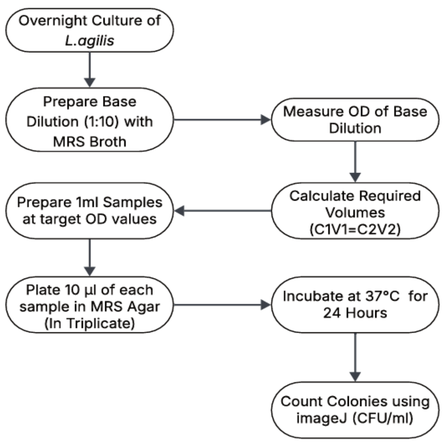

Figure 2: Workflow for standardizing L.agilis viable cell concentration using OD and CFU enumeration

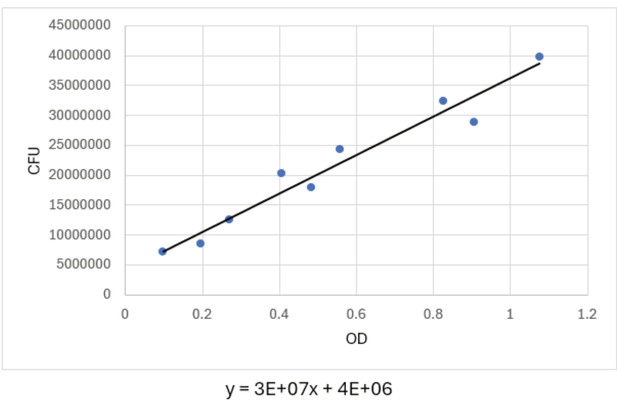

Figure 3: Calibration curve showing a linear relation between OD and CFU

2 Natural dye bioink shows visible, quantifiable response to *L. agilis*

- **Anthocyanin** bioink: probiotic samples showed **pH decrease** and a **blue-shift** in UV-Vis peak with visible hue change over storage
- Confirms metabolic acidification drives the signal
- Supports **refrigerated smart-packaging** use-case

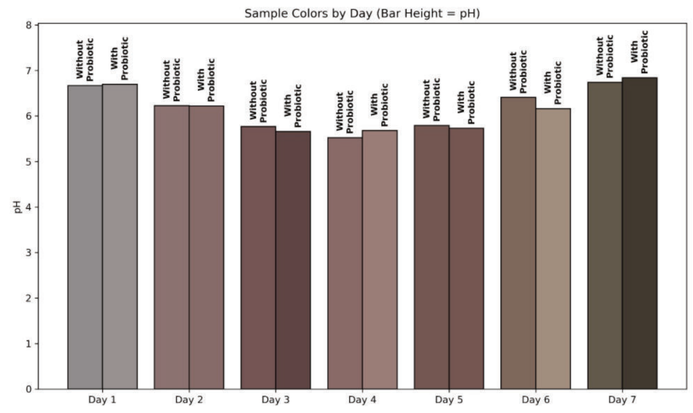

Figure 4: Visual and pH Changes in Anthocyanin-Based Bioink with and Without L. agilis

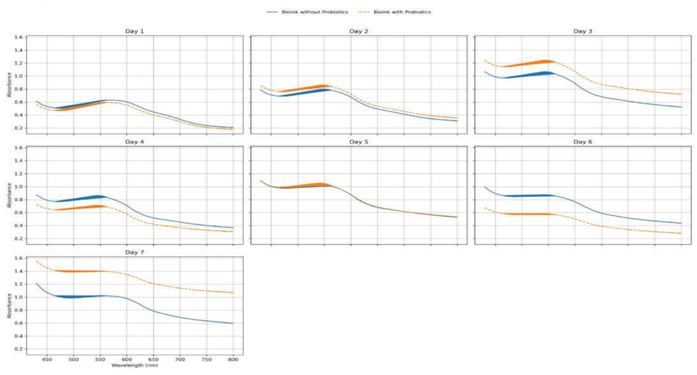

Figure 5: UV–Vis Spectral Changes of Anthocyanin-Based Bioink

3 CRISPR-Cas12a Molecular Confirmation Assay

- **Target:** 16S–23S ISR (species-informative for *L. agilis*).
- **PCR 1:** 137 bp ISR PCR (published) → gel band confirms the right region.
- **PCR 2:** ~780 bp ISR PCR (designed) → clean Sanger reads for reliable sequence.
- **Cross-check:** Align 137 bp and 780 bp (± WGS) → confirm *L. agilis* specificity.
- **Cas12a setup:** Find **TTTV** PAM in confirmed ISR → design **crRNA** at a conserved site.

Figure 6: Agarose gel showing 137 bp amplicon (Lane 1: DNA ladder-50bp)

Species	Target Region	Primer	Sequence (5'→3')	Product Size (bp)
L. agilis	16S–23S ISR	Agilis-F	TCGTAGCTTGCT ACACCGATTG	137
		Agilis-F	CATAATGACCAT GCGATCATCA	

Figure 7: Primer sequences published by Kim et al., 2020, that are used for PCR amplification of the 16S–23S ISR region in Lactobacillus agilis

Key Takeaways

- Bioinks may drift over long storage (Figure 4,5,8) —**stabilizers needed**

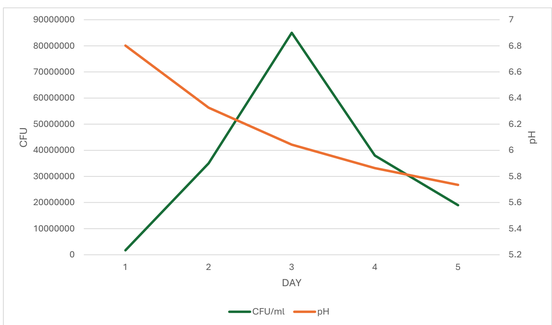

Figure 8: Relationship between pH and CFU/mL over a 5-day period in bioink containing Lactobacillus agilis without indicators

- Short amplicons (137 bp) can challenge Sanger near primer sites; **longer ISR fragments (780 bp)** improve sequence reliability for crRNA design.

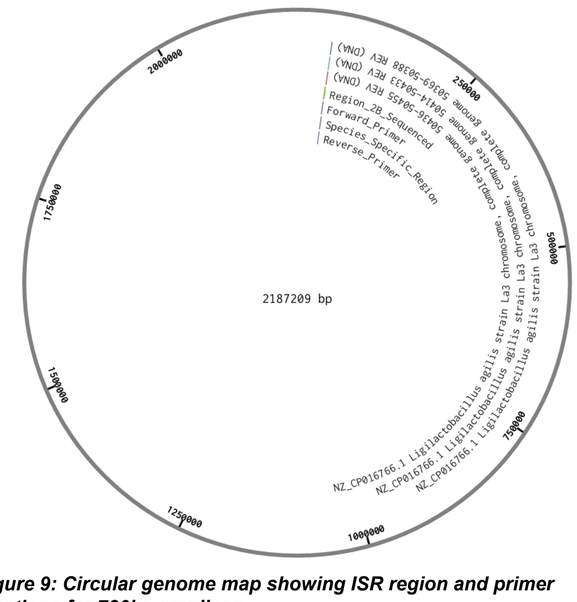

Figure 9: Circular genome map showing ISR region and primer locations for 780bp amplicon

Next Steps

- Cas12a trans-cleavage readout (**Lateral Flow**)

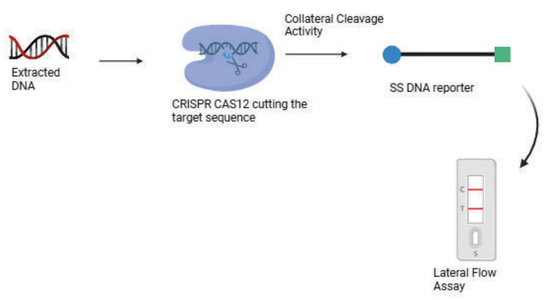

Figure 10: Collateral Cleavage–Driven Lateral Flow Detection (Cas12a)

- **Matrix testing** (yogurt/gut)
- **Multiplexing** to broaden probiotic panels

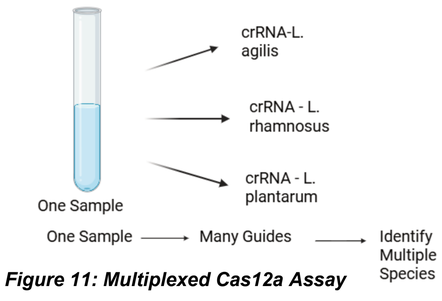

Figure 11: Multiplexed Cas12a Assay

Conclusion

- Reliable Quantification: **OD–CFU calibration curve** (controlled growth) enables consistent inoculum estimates and biosensor fabrication **without repeated plating**.
- **Phenotypic signal: anthocyanin** bioink shows clear, measurable color/pH shifts that track **metabolic activity** of the probiotic (visible, low-cost readout).
- **Molecular confirmation: 137 bp ISR** (published primers) **validated**; an extended **~780 bp ISR** amplicon (in-silico primers) will provide **clean Sanger sequence** for accurate Cas12a **crRNA** design
- **Limitations noted:** experimental-strain **WGS not performed**; definitive sequence confirmation would rely on WGS.

A **proof-of-concept**, **inexpensive**, and **visually interpretable** platform that couples phenotypic sensing with genotypic specificity, supporting rapid detection and monitoring of *L. agilis*

References

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